

Earthquake Vibration Frequency Analyzer

Find dominant vibration modes using eigenvalues

Submitted in partial fulfillment of the requirement for the degree of

Bachelor of Engineering In
Computer Science and Engineering

Submitted By

Poorvi Ganiger	01FE25BCS095
Aditi B Nayak	01FE25BCS091
Mrinalini	01FE25BCS081
Sayee	01FE25BCS089

Under the guidance of
Sumedha Shinde

Academic year 2025-26

Table of Contents

1. Title of Project	01
2. Problem Statement	01
3. Objective	01
4. Mathematical Modeling	1-2
5. Algorithm / Workflow	04
6. Implementation Details	5-9
7. Output Screenshots	9-11
8. Result Analysis	12
9. Conclusion	13
10. References	14

1. Title of the project

Earthquake Vibration Frequency Analyzer – Finding Dominant Seismic Modes Using Eigenvalues and Eigenvectors .

2. Problem Statement

Earthquakes generate vibrations that affect the stability and safety of structures. Understanding the dominant vibration characteristics of earthquake-related parameters is important for identifying significant patterns in seismic behavior.

The objective of this project is to analyze earthquake data and determine the dominant vibration modes using Linear Algebra concepts, specifically Eigenvalues and Eigenvectors. The analysis helps identify which factors contribute most to the overall seismic variation.

3. Objective

The main objectives of this project are:

- To apply Linear Algebra concepts to real-world earthquake data.
- To construct a correlation matrix from important seismic parameters.
- To compute Eigenvalues and Eigenvectors of the correlation matrix.
- To identify dominant vibration modes present in the dataset.
- To determine the contribution of each mode to the overall system behavior.
- To analyze the influence of Magnitude, Significance, and Depth on earthquake vibrations.

4. Mathematical Modeling

The project utilizes three important earthquake parameters:

- Magnitude (M)
- Significance (S)
- Depth (D)

These parameters are represented as a data matrix:

$$A = \begin{bmatrix} M_1 & S_1 & D_1 \\ M_2 & S_2 & D_2 \\ \vdots & \vdots & \vdots \\ M_n & S_n & D_n \end{bmatrix}$$

Correlation Matrix

The relationship among the variables is captured using the correlation matrix:

$$C = A_{corr}$$

For the dataset used, the correlation matrix obtained is:

$$C = \begin{bmatrix} 1.000 & 0.512 & 0.017 \\ 0.512 & 1.000 & -0.079 \\ 0.017 & -0.079 & 1.000 \end{bmatrix}$$

Eigenvalue Decomposition

The vibration modes are obtained by solving:

$$Cv = \lambda v$$

where:

- λ = Eigenvalue
- v = Eigenvector

The Eigenvalues obtained are:

$$\lambda_1 = 0.479$$

$$\lambda_2 = 1.516$$

$$\lambda_3 = 1.005$$

The largest Eigenvalue represents the dominant vibration mode.

Mode Contribution

The contribution of each mode is calculated as:

$$\text{Contribution} = \frac{\lambda_i}{\sum \lambda_i} \times 100$$

5. Algorithm / Workflow

Step 1: Load the earthquake dataset using Pandas.

Step 2 : Select the important parameters:

- Magnitude
- Significance
- Depth

Step 3: Construct the correlation matrix.

Step 4 : Perform Eigenvalue and Eigenvector decomposition using NumPy.

Step 5 : Identify the dominant Eigenvalue.

Step 6 : Classify vibration modes as:

- Dominant vibration mode
- Strong vibration mode
- Weak vibration mode

Step 7 : Compute percentage contribution of each mode.

Step 8 : Analyze the Eigenvector corresponding to the largest Eigenvalue to determine influential factors.

Step 9 : Interpret the results and draw conclusions.

6. Implementation Details

Software Used

Python 3.x

Jupyter Notebook

Libraries Used

```
import pandas as pd
import numpy as np
```

Dataset

Earthquake dataset containing:

- Magnitude
- Significance
- Depth

Key Operations Performed

Loading Dataset

```
df = pd.read_csv("eathquake_finaldataset.csv", skiprows=1)
```

Selecting Features

```
A = df[['magnitude', 'sig', 'depth']]
```

Correlation Matrix

```
M = A.corr()
```

Eigenvalue Decomposition

```
vals, vecs = np.linalg.eig(M)
```

Dominant Mode Detection

```
dominant = np.argmax(vals)
```

Mode Contribution Calculation

```
contribution = (v/total)*100
```

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("eathquake_finaldataset.csv", skiprows=1)

print("Dataset Shape:", df.shape)
df.head()
```

```
[3]: A = df[['magnitude', 'sig', 'depth']]

print("Selected Features:")
print(A.head())
```

```
[4]: M = A.corr()

print("Correlation Matrix:")
print(M)
```

```
[5]: vals, vecs = np.linalg.eig(M)

print("Eigenvalues:")
print(vals)

print("\nEigenvectors:")
print(vecs)
```

```
[6]: print("System Behaviour Analysis\n")

for i, v in enumerate(vals):

    if v == max(vals):
        status = "Dominant vibration mode"

    elif v > 1:
        status = "Strong vibration mode"

    else:
        status = "Weak vibration mode"

    print(f"Mode {i+1} → {v:.3f} : {status}")
```

```
[7]: total = np.sum(vals)

print("Mode Contributions\n")

for i, v in enumerate(vals):

    contribution = (v / total) * 100

    print(f"Mode {i+1} : {contribution:.2f} %")
```

```
[8]: dominant = np.argmax(vals)

print("Dominant Mode Factor Weights\n")

features = ['Magnitude', 'Significance', 'Depth']

for f, w in zip(features, vecs[:, dominant]):

    print(f"{f:12} : {w:.3f}")
```



```
[9]: plt.figure(figsize=(6,4))

plt.bar(
    ['Mode 1', 'Mode 2', 'Mode 3'],
    vals
)

plt.title('Dominant Seismic Modes')
plt.xlabel('Modes')
plt.ylabel('Eigenvalue')

plt.show()
```

```
[10]: contributions = (vals / np.sum(vals)) * 100

plt.figure(figsize=(6,4))

plt.bar(
    ['Mode 1', 'Mode 2', 'Mode 3'],
    contributions
)

plt.title('Mode Contribution Percentage')
plt.xlabel('Modes')
plt.ylabel('Contribution (%)')

plt.show()
```

7. Output Screenshots

Dataset Shape: (1000, 10)

[2]:	magnitude	date_time	alert	sig	nst	gap	depth	latitude	longitude	location
0	6.5	16-08-2023 12:47	green	657	114	25.0	192.955	-13.8814	167.1580	Sola, Vanuatu
1	6.5	19-07-2023 00:22	yellow	775	92	40.0	69.727	12.8140	-88.1265	Intipucá, El Salvador
2	6.6	17-07-2023 03:05	green	899	70	28.0	171.371	-38.1911	-70.3731	Loncopué, Argentina
3	7.2	16-07-2023 06:48	green	860	173	36.0	32.571	54.3844	-160.6990	Sand Point, Alaska
4	7.3	16-07-2023 06:48	NaN	820	79	172.8	21.000	54.4900	-160.7960	Alaska Peninsula

Selected Features:

	magnitude	sig	depth
0	6.5	657	192.955
1	6.5	775	69.727
2	6.6	899	171.371
3	7.2	860	32.571
4	7.3	820	21.000

Eigenvalues:

[0.47870681 1.51613637 1.00515683]

Eigenvectors:

[[0.69718 -0.70048835 0.15249955]
 [-0.70500283 -0.70850904 -0.03139978]
 [-0.13004249 0.08562132 0.98780461]]

Correlation Matrix:

	magnitude	sig	depth
magnitude	1.000000	0.512358	0.017083
sig	0.512358	1.000000	-0.079263
depth	0.017083	-0.079263	1.000000

System Behaviour Analysis

Mode 1 → 0.479 : Weak vibration mode

Mode 2 → 1.516 : Dominant vibration mode

Mode 3 → 1.005 : Strong vibration mode

Mode Contributions

Mode 1 : 15.96 %

Mode 2 : 50.54 %

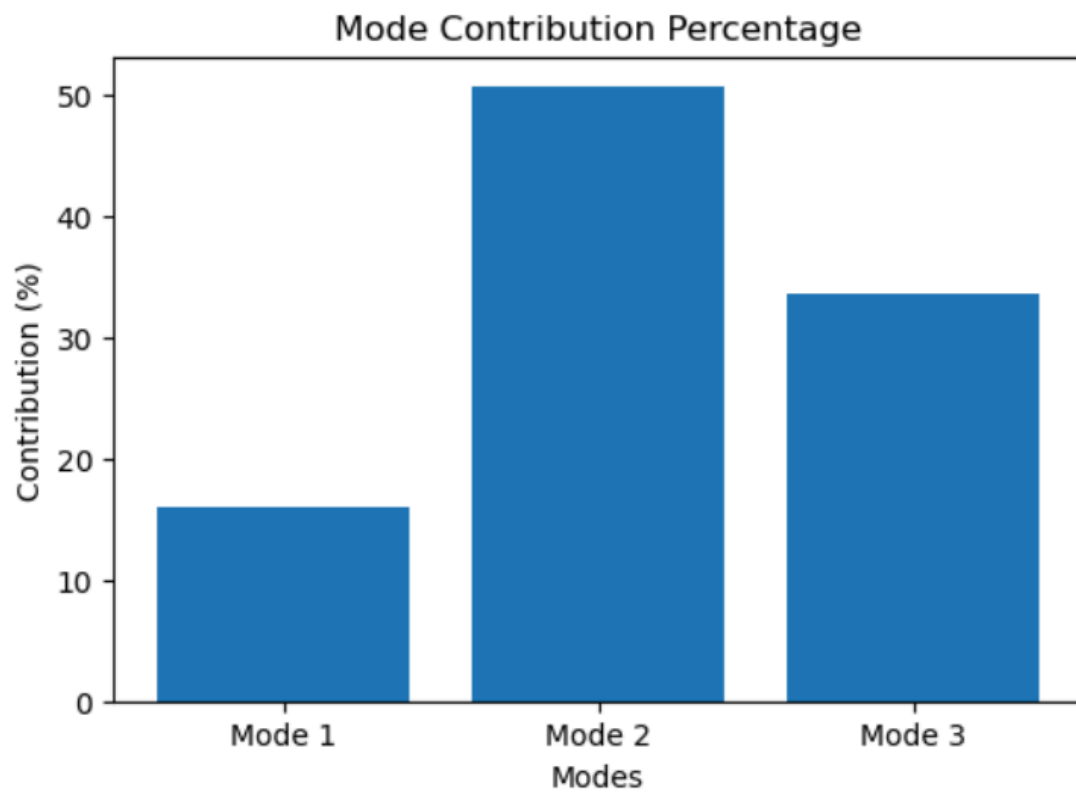
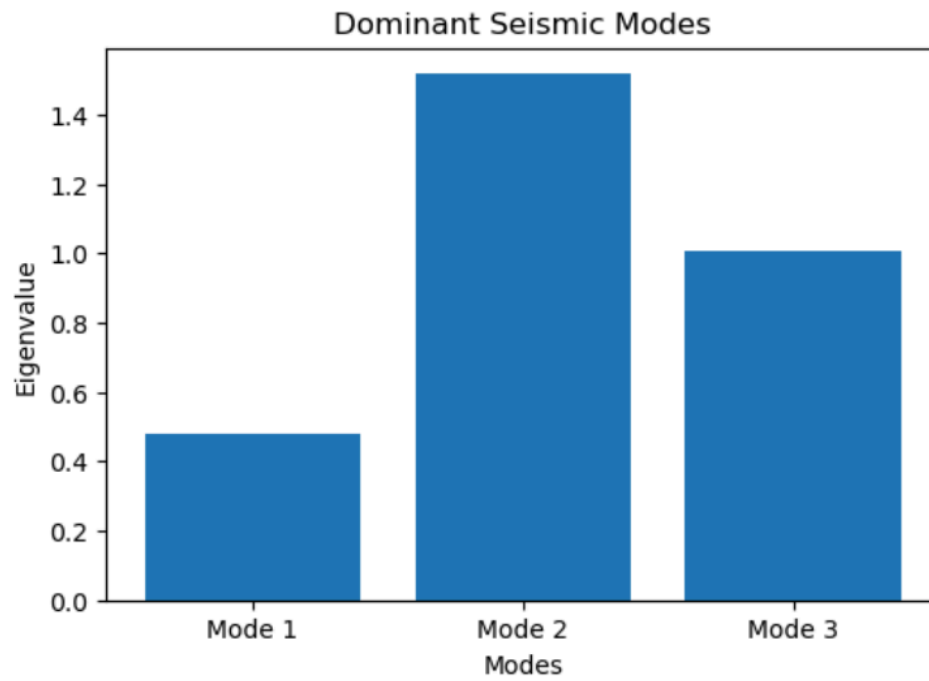
Mode 3 : 33.51 %

Dominant Mode Factor Weights

Magnitude : -0.700

Significance : -0.709

Depth : 0.086



8. Result Analysis

The correlation matrix indicates a moderate positive relationship between Magnitude and Significance (0.512), while Depth shows very weak correlation with both variables.

The Eigenvalue decomposition produced three vibration modes:

Mode	Eigenvalue	Interpretation
Mode 1	0.479	Weak vibration mode
Mode 2	1.516	Dominant vibration mode
Mode 3	1.005	Strong vibration mode

The dominant Eigenvalue (1.516) contributes approximately **50.54%** of the total system variation, indicating that this mode captures most of the seismic behavior.

The corresponding Eigenvector is:

$$[-0.700, -0.709, 0.086]$$

This shows that:

- Magnitude has a strong influence on the dominant mode.
- Significance has a strong influence on the dominant mode.
- Depth has very little influence compared to the other two factors.

Thus, earthquake vibration characteristics are primarily governed by Magnitude and Significance.

9. Conclusion

This project successfully applied Linear Algebra concepts to analyze earthquake vibration characteristics. By constructing a correlation matrix and performing Eigenvalue-Eigenvector decomposition, dominant vibration modes were identified.

The analysis revealed that:

- The second mode is the dominant vibration mode.
- It contributes more than 50% of the total system variation.
- Magnitude and Significance are the most influential factors affecting seismic vibrations.
- Depth contributes relatively less to the dominant vibration pattern.

The project demonstrates the practical application of Eigenvalues and Eigenvectors in extracting meaningful insights from real-world earthquake datasets and understanding vibration behavior.

10. References

- NumPy Documentation – Eigenvalue Decomposition.
- Pandas Documentation.
- Linear Algebra and Its Applications – Gilbert Strang.
- Earthquake Dataset used for analysis (Kaggle Dataset).
- Python Official Documentation.

